THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A body of mass 2 kg is thrown vertically upward with 20 m/s. The KE at the top is: 1

(a) 200 J **(b)** 0 **(c)** 400 J **(d)** 40 J

2. An engine of 2 kW works for 2 hours. The work done is: 1

(a) 1.44 x 10^6 J **(b)** 14.4 x 10^6 J **(c)** 7.2 x 10^6 J **(d)** 4 x 10^3 J

3. A pulley system lifts 200 N with an effort of 50 N over 1 m of load while effort moves 5 m. The efficiency is: 1

(a) 20 per cent **(b)** 50 per cent **(c)** 80 per cent **(d)** 100 per cent

4. A hydroelectric dam water of 1000 kg falls through 50 m. The maximum energy that can be obtained is (g = 10): 1

(a) 50 J **(b)** 50000 J **(c)** 500000 J **(d)** 5000 J

5. A 2 kg stone is at 10 m. When it has fallen 4 m its kinetic energy is (g = 10): 1

(a) 80 J **(b)** 120 J **(c)** 40 J **(d)** 200 J

6. Two persons climb a staircase: A in 10 s and B in 20 s. A has: 1

(a) half the power of B **(b)** equal power

(c) no power **(d)** twice the power of B

7. An ideal machine has an efficiency of: 1

(a) 100 per cent **(b)** 0

(c) more than 100 per cent **(d)** 50 per cent

8. A roller coaster car starts from rest at a height of 20 m. At a height of 5 m the speed is (g = 10): 1

(a) 17.3 m/s **(b)** 20 m/s **(c)** 30 m/s **(d)** 10 m/s

9. A 2 kg stone dropped from 10 m. Its speed at 6 m above ground is: 1

(a) 8.9 m/s **(b)** 14 m/s **(c)** 10 m/s **(d)** 4.5 m/s

10. The power of an engine that exerts a force of 500 N at a speed of 10 m/s is: 1

(a) 490 W **(b)** 5000 W **(c)** 510 W **(d)** 50 W

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A machine lifts 400 N through 2 m with an effort of 250 N moving 4 m. Find the efficiency. 2

12. An electric motor of 1 kW works for 30 minutes. Find the energy in kWh and in joules. 2

13. A man pushes a box 5 m with 20 N of force and friction 8 N opposes. Find the work done by him and by friction. 2

14. State why a nut is easier to loosen with a long spanner. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bulb of 40 W is replaced by an LED of 8 W for 5 hours a day. Find the energy saved in a 30-day month and the saving at 6 rupees per unit. 3

16. A man of mass 60 kg climbs a hill of height 100 m in 10 minutes. Find his power in watts and in horsepower (1 hp = 746 W). 3

17. Explain how a see-saw works with two children of masses 30 kg and 40 kg and the position needed for balance if the lighter child sits 2 m from the pivot. 3

OR

Explain why a nail is driven by a hammer easily but pushing the nail by hand does not drive it in, in terms of KE and work. 3

18. Calculate the energy and the cost: a 2 kW geyser works 30 minutes a day for 30 days at 6 rupees per unit. 3

OR

A wheel and axle has a wheel radius of 20 cm and axle radius of 5 cm. Find the MA, the effort for a 200 N load. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the energy conversions along the path of a roller-coaster car with a track: start at 30 m, valley at 0 m, hill of 20 m, valley again; compute the speeds at the valley and at the hill (g = 10). 5

OR

Explain the energy changes when a boy jumps on a trampoline from the top of a jump to the lowest point, including elastic PE and heat. 5

20. A block of mass 5 kg is pushed up a smooth incline of length 10 m and height 4 m. Find the force needed, the work done and the PE gained (g = 10). 5

OR

Compare a human, a motor and an engine in power with 100 W, 1 kW and 100 kW, and calculate the time each needs to lift 1000 kg through 10 m. 5

END OF QUESTION PAPER — Total: 40 Marks

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